DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
ABES ENGINEERING COLLEGE, GHAZIABAD

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title: Incident Detection for a DNS Tunneling Attack (P_210)
Standard Academic Compliance: IEEE Std 830-1998

SUBMITTED BY:
Harsh Juneja (Roll No: 2400320100487)
Degree: Bachelor of Technology (B.Tech)

UNDER THE GUIDANCE OF:
Mr. Priyan S
Department of Computer Science & Engineering

Status: 3rd Year Major Project Specification
Project Code: P_210

Academic Year: 2026 - 2027
Date of Submission: 31-10-2026 (Revision 2.0)

Document Revision & Approval History

Version	Date	Description / Major Changes	Prepared By	Reviewed By
0.1	15-Sep-2026	Initial Draft & Scope Definition	Harsh Juneja	Mr. Priyan S
1.0	30-Sep-2026	Final 3rd Year Mid-Term SRS Baseline	Harsh Juneja	Mr. Priyan S
2.0	31-Oct-2026	Major Production Upgrade: Live Dashboard Sniffing UI, Cloud Traffic Simulation Stream, PCAP Parser, Heuristic Policy Dataclass, Threat Severity Tiers (Critical/High/Med/Low), 1-Click CSV Exports, Corporate Light Theme, 1-Click Desktop Launcher, and Automated GitHub Actions CI/CD Pipeline.	Harsh Juneja	Mr. Priyan S

Guidance Note: *This Software Requirements Specification document conforms strictly to the IEEE Std 830-1998 recommended practice. All functional requirements, operational environments, interface specifications, and behavioral workflows reflect the updated production baseline for 3rd-year university evaluation, SOC lab defense, and project grading.*

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, clear, and unambiguous definition of the functionality, external interfaces, performance targets, and constraints of the proposed system: 'Incident Detection for a DNS Tunneling Attack (P_210)'. This specification serves as a binding engineering reference between student developers, project evaluators, academic supervisors, and prospective end users during the 3rd-year engineering curriculum and industry project-based learning assessments.

1.2 Scope of the System

The proposed application is designed to identify and triage suspicious DNS tunneling and covert data exfiltration patterns by delivering an end-to-end detection, analysis, and monitoring workflow. The system has evolved from a local script-driven prototype into an enterprise-grade Security Operations Center (SOC) dashboard. Key boundary guidelines include:

- In-Scope Capabilities: Multi-vector DNS telemetry ingestion (synthetic benchmark datasets, real-time local network packet sniffing via Scapy with safe timeouts, cloud-friendly live traffic simulation streams, and Wireshark PCAP/PCAPNG file uploads); mathematical feature extraction (Shannon entropy, domain length, subdomain depth, query frequency, and record types); centralized heuristic risk scoring engine (0–7 scale) with configurable policy constants; threat severity categorization (CRITICAL, HIGH, MEDIUM, LOW); explainable alert generation with human-readable detection rationale; real-time SOC analytical dashboard visualization in a formal corporate light theme; one-click CSV report export for incident alerts and traffic logs; 1-click Windows desktop launcher; and automated GitHub Actions CI/CD test verification.
- Out-of-Scope Boundaries: Automatic firewall kernel blocking or active host shutdown (to prevent accidental enterprise service disruption); complex multi-tenant enterprise SIEM installations; relational database backends (CSV and in-memory session states are utilized for maximum portability and rapid forensic review); live malware detonation; and direct decryption of encrypted transport layers (DoH/DoT) in the current prototype release.
- Expected Benefits: Safe and repeatable academic testing; improved forensic visibility into Port 53 DNS query payloads; explainable risk-scored alerting to alleviate SOC analyst alert fatigue; zero-configuration deployment via Streamlit Community Cloud; and immediate 1-click execution for university evaluators.

1.3 Definitions, Acronyms, and Abbreviations

Category	Term / Acronym	Definition / Standard Context
Standard	SRS	Software Requirements Specification (IEEE Std 830-1998 standard format).
Network	DNS	Domain Name System; standard hierarchical protocol operating over UDP/TCP Port 53.
Network	QNAME	DNS query name; contains the fully qualified domain name (FQDN) requested by the client.
Network	QTYPE	DNS query record type, such as A (IPv4), AAAA (IPv6), TXT, MX, or CNAME.
Security	DNS Tunneling	Covert cyberattack technique weaponizing DNS queries/responses to encapsulate unauthorized C2 instructions or exfiltrate sensitive data.
Security	Entropy	Shannon entropy measure calculating character randomness; values > 3.5 bits/char identify encoded/encrypted Base32/Base64 payloads.

Security	Severity Tiers	Standardized SOC triage classification: CRITICAL (6-7), HIGH (4-5), MEDIUM (2-3), and LOW (0-1).
Security	MITRE ATT&CK	Globally accessible knowledge base of adversary tactics; maps to T1071.004 (DNS C2) and T1041 (Exfiltration).
Project Specific	Scapy	Powerful Python packet manipulation and network sniffing engine utilized for Port 53 socket interception.
Project Specific	PCAP / PCAPNG	Packet Capture file format generated by network analyzers such as Wireshark and tcpdump.
Project Specific	Streamlit	Python application framework powering the interactive SOC monitoring dashboard.
Project Specific	Risk Score	Granular composite heuristic score (0–7) derived from weighted evaluation of 5 independent telemetry features.
Project Specific	CI/CD	Continuous Integration / Continuous Deployment; automated GitHub Actions test execution matrix on Python 3.10 and 3.11.
Project Specific	Streamlit Cloud	Serverless public cloud hosting platform hosting the live deployed application.

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- RFC 1035: Domain Names - Implementation and Specification (Mockapetris, P.).
- NIST Special Publication 800-81r3: Secure Domain Name System (DNS) Deployment Guide.
- MITRE ATT&CK Framework: Technique T1071.004 (Application Layer Protocol: DNS) & Technique T1041 (Exfiltration Over C2 Channel).
- Scapy Documentation: Packet Sniffing, Promiscuous Socket Filtering, and Dissection.
- Streamlit Documentation: Session State Management, Layout Theming, and Data Visualization.

1.5 Document Overview

The remainder of this specification is partitioned into structural IEEE sections: Section 2 outlines the product perspective, target user classes, operational context, and updated design constraints. Section 3 specifies detailed functional requirements partitioned by subsystem modules with unique requirement IDs. Section 4 defines external user, hardware, software, and communication interfaces. Section 5 articulates non-functional attributes (Performance, Reliability, Security). Section 6 presents academic system design and behavioral models. Finally, Section 7 provides university review and approval sign-off.

2. Overall Description

2.1 Product Perspective

The Incident Detection System (P_210) is architected as an autonomous, self-contained Python cybersecurity platform. It bridges network telemetry capture, mathematical feature extraction, explainable heuristic inference, and web-based SOC visualization. Unlike monolithic enterprise SIEMs requiring heavy infrastructure, this system provides modular zero-friction execution locally via Windows/Linux environments and remotely via Streamlit Community Cloud.

2.2 User Classes and Characteristics

User Role / Persona	Technical Proficiency	Core Responsibilities & Rights
SOC Analyst / SME	Intermediate to High	Monitor real-time DNS telemetry, evaluate risk scores and detection rationale, triage alerts, download incident CSV reports, and initiate quarantine procedures.
Developer / Student	Intermediate	Maintain codebase, calibrate heuristic policy thresholds in config.py, run automated unit test suites, trigger packet simulation pipelines, and push updates.
Academic Evaluator	Basic to Intermediate	Review project compliance against IEEE standards, evaluate live demonstration via web link, inspect test results, and assess scoring accuracy.

2.3 Operating Environment

- Client Interface: Modern HTML5 web browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari) accessing localhost:8501 or the deployed cloud URL.
- Local Host Environment: Microsoft Windows 10/11 or Ubuntu Linux running Python 3.10, 3.11, or 3.12 with an isolated virtual environment (venv).
- Cloud Hosting Environment: Streamlit Community Cloud (Linux container) continuously synced with the GitHub repository main branch.
- Software Libraries: Pandas, Streamlit, Scapy, Openpyxl, and Python-docx.
- Packet Sniffing Support: Local live NIC packet capture utilizes Scapy, requiring Npcap (on Windows) or libpcap with raw socket permissions.
- Automated CI/CD Environment: GitHub Actions runners running automated matrix tests on Ubuntu-latest.

2.4 Design and Implementation Constraints

- Browser Sandboxing Constraint: Web browsers strictly prevent web applications from sniffing raw network interfaces on a visitor's PC. The updated design resolves this constraint by providing: (1) Local Scapy sniffing with safe timeouts, (2) One-click live cloud traffic stream simulation, and (3) Client-side PCAP file upload.
- Tunable Heuristic Constraint: Detection parameters are decoupled into detection/config.py (DetectionConfig dataclass) rather than hardcoded magic constants, allowing dynamic calibration.

- Explainability Constraint: The inference engine must remain 100% explainable; black-box ML models are excluded to provide clear audit trails for SOC triage.
- Execution Performance Constraint: Live packet sniffing enforces configurable timeouts (3s–20s) and packet sliders (10–100 packets) to prevent dashboard thread blocking.

2.5 Assumptions and Dependencies

- Network Visibility: For local live capture, the host machine must have active UDP/TCP Port 53 DNS traffic passing through the selected network interface.
- Dependency Integrity: Python packages specified in requirements.txt (pandas, streamlit, scapy) must be installed.
- Safe Sandbox Guarantee: Synthetic datasets and simulation generators represent controlled, safe educational data and contain no weaponized exploit payloads.

3. System Features & Functional Requirements

3.1 Module 1: Telemetry Ingestion & Packet Processing

Req ID	Feature Description	Input / Validation Rule	Priority
FR-1.1	Synthetic Dataset Ingestion	System shall read pre-generated CSV datasets (dns_traffic.csv) containing benign and suspicious DNS query records.	High
FR-1.2	Live Network Sniffing	System shall capture live UDP/TCP Port 53 DNS packets from physical NICs using Scapy with configurable packet counts (10–100) and timeouts (3–20s).	High
FR-1.3	Cloud Traffic Simulation Stream	System shall generate a live burst of realistic DNS queries (benign enterprise domains + tunneling attacks) to enable interactive live testing on deployed cloud servers without raw socket privileges.	High
FR-1.4	PCAP Capture File Upload	System shall allow users to upload .pcap and .pcapng files captured from Wireshark, parsing DNS records into tabular telemetry for offline inspection.	Medium

3.2 Module 2: Feature Extraction & Detection Engine

Req ID	Feature Description	Input / Validation Rule	Priority
FR-2.1	Mathematical Feature Extraction	System shall extract domain length, subdomain count, Shannon entropy, query frequency, and query type for every query record.	High
FR-2.2	Shannon Entropy	System shall compute entropy using the formula:	High

	Computation	H = -sum(p * log2(p)). Queries with entropy > 3.5 bits/char shall be flagged as high-entropy payloads.	
FR-2.3	Heuristic Rule Evaluation	System shall apply weighted scoring: Length >50 (+2), Subdomains >=2 (+1), Entropy >3.5 (+2), Frequency >=5 (+1), and TXT query (+1).	High
FR-2.4	Tunable Configuration Constants	System shall encapsulate detection policy thresholds in a DetectionConfig dataclass, supporting runtime customization without altering source logic.	Medium
FR-2.5	Threat Severity Classification	System shall categorize risk scores (0–7) into standard SOC severity tiers: CRITICAL (6–7), HIGH (4–5), MEDIUM (2–3), and LOW (0–1).	High

3.3 Module 3: SOC Visualization, Alerting & Reporting

Req ID	Feature Description	Input / Validation Rule	Priority
FR-3.1	Enterprise SOC Dashboard	System shall render a formal corporate light-themed dashboard with real-time KPI metric cards, status alert banners, and interactive controls.	High
FR-3.2	Interactive Mode Switching	System shall toggle seamlessly between Live DNS Monitoring and Synthetic Dataset Testing modes via responsive radio selectors.	High
FR-3.3	Threat Distribution Analytics	System shall render interactive bar charts visualizing suspicious query frequencies by source IP and risk score breakdown.	Medium
FR-3.4	One-Click Incident CSV Export	System shall provide direct download buttons allowing analysts to export raw DNS traffic logs and categorized incident reports as CSV files.	High
FR-3.5	1-Click Desktop Launcher	System shall provide a Windows batch launcher (run_dashboard.bat) enabling 1-click execution without terminal command entry.	Medium

3.4 Module 4: Verification, Testing & CI/CD Pipeline

Req ID	Feature Description	Input / Validation Rule	Priority
FR-4.1	Automated Unit Test Suite	System shall maintain an automated test suite (tests/) covering entropy math, domain parsing, heuristic scoring, and packet pipelines.	High

FR-4.2	GitHub Actions CI Matrix	System shall trigger automated CI workflows on push/PR to main, testing test execution across Python 3.10 and 3.11 with zero failures.	High
FR-4.3	Zero-Regression Baseline	All 11 unit tests shall execute and pass in under 0.05 seconds prior to any release or commit deployment.	High

4. External Interface Requirements

4.1 User Interfaces (UI)

The interface adheres to formal executive SOC ergonomics utilizing a corporate light theme: background #F8FAFC, crisp white container cards #FFFFFF with subtle #E2E8F0 borders, and high-contrast charcoal #0F172A typography. The dashboard presents: (1) Enterprise Header with status badges; (2) Interactive Live Capture Controls with tabs for Local Sniffing, Cloud Traffic Simulation, and PCAP Upload; (3) 4 High-Impact KPI Metric Cards; (4) Incident Alert Banner; (5) Filterable DNS Traffic and Incident Alert Tables; (6) Export Actions; and (7) Source IP & Risk Distribution charts.

4.2 Hardware Interfaces

Standard PC hardware: Display resolution >= 1280x720; Keyboard and Mouse; and a standard network interface card (NIC) (Ethernet or 802.11 Wi-Fi) supporting promiscuous socket capture via Scapy/Npcap for local live testing.

4.3 Software Interfaces

- Operating System: Windows 10/11 or Ubuntu Linux.
- Runtime: Python 3.10 / 3.11 / 3.12.
- Network Sniffing: Scapy 2.6+ with Npcap.
- Analytical Layer: Pandas 2.2+.
- Frontend Framework: Streamlit 1.39+.
- Continuous Integration: GitHub Actions (Ubuntu-latest runners).

4.4 Communication Interfaces

- Web Transport: HTTP/HTTPS over Port 8501 locally, and standard Port 443 on Streamlit Community Cloud.
- DNS Packet Telemetry: Ingestion filtered for UDP Port 53 and TCP Port 53.
- Data Persistence: Tabular CSV file streaming.

5. Non-Functional Requirements (NFRs)

Attribute	Target Metric / Standard	Verification & Testing Method
5.1 Performance	Feature calculation and risk evaluation shall complete in < 30ms for 100 queries. Live packet capture timeout enforced between 3s–20s to ensure zero	Benchmarked via automated test execution (11 tests pass in 0.025s); verified on Streamlit Community Cloud.

	dashboard UI freeze.	
5.2 Security & Containment	System operates strictly as a read-only telemetry sensor and analytical alerting tool; does not execute captured payloads or disrupt corporate DNS traffic.	Code review verifying read-only parsing and safe string sanitization on decoded QNAME packets.
5.3 Reliability & Correctness	100% deterministic mathematical calculations for Shannon entropy and heuristic scoring for identical inputs.	Automated unit tests (test_feature_extractor, test_detection_engine) executing via GitHub Actions CI on every commit.
5.4 Portability & Usability	Zero-install access for evaluators via public cloud URL; 1-click execution on local Windows machines via run_dashboard.bat.	Verified multi-environment execution across local Windows 11 and cloud Linux container deployment.

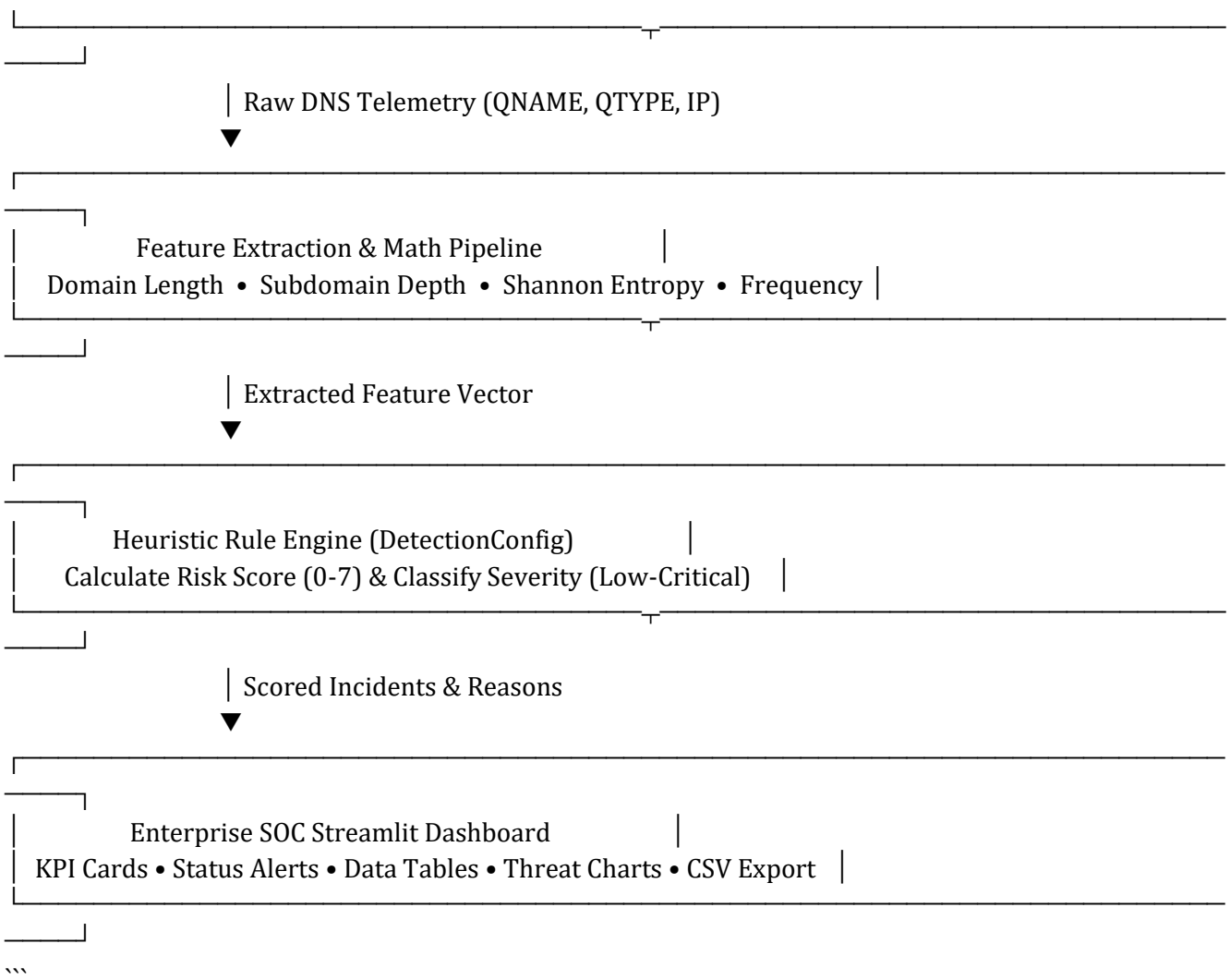
6. System Design and Analysis Models

In accordance with IEEE 830-1998 guidelines and academic evaluation rubrics, the system architecture and dynamic behavior are modeled below:

Diagram Model	Objective & Operational Scope	Deliverable Architecture
System Context Diagram	Illustrates external data boundaries (Network NIC, Synthetic Data, PCAP Files, Cloud Simulator) interacting with the Detection System Boundary and SOC Dashboard.	Multi-source Ingestion -> Detection Core -> Streamlit Web Dashboard.
Data Flow Diagram (Level 0/1)	Depicts telemetry pipeline: Raw Packets -> Feature Extraction (Entropy, Length, Depth) -> Heuristic Scoring Engine -> Alerts & CSV Persistence -> Dashboard.	Modular Data Pipeline Flow.
Use Case Diagram	Maps actor interactions (SOC Analyst, Student Developer, Academic Evaluator) across Live Capture, Cloud Simulation, PCAP Upload, Alert Review, and CSV Export.	Comprehensive UML Actor-Interaction Model.
Sequence Diagram	Chronological message exchanges between User, Streamlit Frontend, Scapy/Simulation Engine, Detection Core, and CSV Export Controllers.	End-to-end interactive sequence workflow.

6.1 Architectural Telemetry Flow

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                  Multi-Vector Ingestion Layer                                        │
│  [Local Scapy Sniffer]  [Cloud Simulation Stream]  [PCAP File Upload]                                │
```



7. Academic Review & Sign-Off

This updated Software Requirements Specification (SRS v2.0) has been formally submitted for evaluation and project review by the Department of Computer Science & Engineering, ABES Engineering College:

Project Coordinator / Guide	Internal Examiner	Head of Department (HOD)
_____	_____	_____
Mr. Priyan S	Department Panel Member	Computer Science & Engineering
Date: 31-10-2026	Date: 31-10-2026	Date: 31-10-2026